Md. Hasibul Islam

+8801799885464 Email Website github linkedin Dhaka,Bangladesh

Education

- **B.Sc. in Computer Science & Engineering (CSE)** Apr 2023 – Jan 2027
  American International University-Bangladesh
  CGPA: 3.75/4.00
- **Higher Secondary Certificate (HSC)** Passing year 2021
  Madaripur Govt. College
  GPA: 5.00/5.00
- **Secondary School Certificate (SSC)** Passing year 2019
  U.I Govt. High School
  GPA: 5.00/5.00

Research Interests

Human-Computer Interaction (HCI), Human Centered AI (HAI), Respoonsible AI, Social Computing, Explainable AI, Computer Vision, Medical Imaging, Deep Learning

Publications

MIME: Multistrategy Imbalance Mitigation Engine for Skin Cancer Classification link

- Presented at International Conference on Trends in Computational and Cognitive Engineering(TCCE 2025), Melaka, Malaysia
- Proposed novel approach to handling class imbalance in dermatological AI systems

Preprint

Reducing Notification Fatigue using a Portable Adaptive-Baseline PM2.5 Monitor for Persistently Polluted Cities (Submitted on 29th ICCIT 2026) link

- Developed a portable, adaptive PM2.5 monitoring system that detects statistically significant pollution spikes instead of relying on fixed thresholds.
- A two-week field study in Dhaka showed that adaptive spike-based alerts substantially reduced notification frequency and improved the meaningfulness of alerts, helping address notification fatigue.

Federated Deep Learning for Alzheimer's Disease Detection: Strong Internal Performance Does Not Survive a Change of Imaging Source (Manuscript prepared for submission to a Springer Nature Journal) link

- A dual-channel ConvNeXt-Small + Swin-V2-S classifier with CBAM fusion, trained centrally and with simulated federated learning (FedAvg, FedProx, FedBN). It reaches 92.74% internal accuracy, but accuracy falls to 12–22% on 86,437 external OASIS-1-derived images. Strong same-source results did not show that the model is robust across imaging sources.

ClinDiffXAI: Concept-Conditioned Diffusion for Clinically Grounded Dermoscopic Image Synthesis ((Manuscript prepared for submission to a Springer Nature Journal)

- Developed ClinDiffXAI, a concept-conditioned diffusion model for clinically controllable dermoscopic image synthesis, achieving statistically significant gains in melanoma feature fidelity over class-conditioned baselines

Experience

Research Assistant (RA) Jun 2025 – Aug 2026
Arche Intelligence Lab

- Build and evaluate different custom CNN models
- Published a research paper in an international conference(TCCE 2025)

Projects

Academic Projects

Eventique – Event Management System (JavaScript, PHP, HTML, CSS) GitHub

- Built a website for an event management system where users can book a hall for their event.
- Signup and signin required before booking
- Admin can maintain(add, delete, update) the whole system.
- Connected the project with a database and followed the **MVC** pattern.

Rural and Urban Scenes – OpenGL project (C++, OpenGL) GitHub

- Built a computer graphics project with different scenes using C++ and OpenGL.
- Implemented day-night changes, dynamic object motion, and direction changes.

TaxiCab - Taxi service App Java

- Built a desktop app for a taxi service system.
- Users can book a taxi for a ride; admin can maintain the whole system.

Netspace - ISP Management System .NET, C#

- Built a .NET desktop app for an ISP company.
- Users can book a package and upgrade their current package.
- Admin can add any user, book a package for them, and monitor the whole system.

Volunteering, Leadership & Extracurriculars

- AIUB Research and Development Club – General Member Dec 2023 – Current
- Bangladesh Scouts – Leadership, teamwork, problem-solving 2014 – 2018
- BD Clean (General Member) 2019 – 2021
  volunteer-based youth organization, working to establish Bangladesh as a clean country

Awards & Achievements

- Dean's List Honor - Faculty of Science and Technology
- Academic Scholarship at AIUB: 60% merit-based

Skills

- Programming: C, C++, Python, R, Java, C#
- Data Analysis: Numpy, Pandas, Matplotlib, Seaborn, Plotly
- Machine Learning: TensorFlow, PyTorch
- Tools: Git, LaTeX

Personal References

Md Kishor Morol
AI Safety Researcher
META
Founder & CEO
ELITE Research Lab LLC
kishor.aiub@gmail.com

Dr. Muhammad Firoz Mridha
Professor
Head [Undergraduate Program]
Department of Computer Science
American International University-Bangladesh
firoz.mridha@aiub.edu